

LAGUNA STATE POLYTECHNIC UNIVERSITY

Bachelor of Science in Information Technology

Enterprise Infrastructure Plan

ABC Startup Solutions

Week 2 Portfolio Project

Course	ITEP 414 - System Administration and Maintenance
Project Type	Individual Portfolio Project
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Executive Summary

ABC Startup Solutions is a newly established software development company with 20 employees working from one office floor. The company currently has no computers, server, network, internet infrastructure, or security policies. This plan establishes a practical starting infrastructure that supports development, human resources, finance, and sales while allowing the company to expand.

The recommended design uses business-class endpoint computers, a centralized Ubuntu Server, a managed 48-port switch, router and firewall, wireless access, NAS storage, UPS protection, structured CAT6 cabling, endpoint security, backups, and documented administrative procedures. The network is designed so that departmental traffic can later be separated using VLANs and access-control rules.

PART 1 – Company Profile

1.1 Company Name

ABC Startup Solutions

1.2 Nature of Business

ABC Startup Solutions is a small software development and technology services company. Its primary activities include web and application development, software maintenance, technical consulting, and support services for business clients.

1.3 Vision

To become a trusted and innovative technology partner that delivers secure, reliable, and practical digital solutions for organizations in the Philippines and beyond.

1.4 Mission

To provide quality software and technology services through skilled people, dependable infrastructure, secure information systems, continuous learning, and customer-focused support.

1.5 Office Location

Fictional Location: 3rd Floor, Laguna Technology Center, National Highway, Santa Rosa, Laguna, Philippines.

1.6 Organizational Structure

Level/Unit	Function
General Manager / Executive Management	Business strategy, approvals, budgets, and overall direction
Information Technology	Software development, infrastructure, user support, and technical operations
Human Resources	Recruitment, employee records, policies, and personnel administration
Finance	Accounting, budgeting, payroll coordination, and financial reporting
Sales	Customer acquisition, account management, proposals, and sales operations

1.7 Employee Distribution

Department	Employees	Primary IT Needs
Information Technology	5	Development workstations, server access, Git, IDEs, technical tools
Human Resources	4	Office productivity, employee records, communication, printing
Finance	5	Office productivity, accounting/financial applications, secure file access
Sales	6	CRM/web applications, office productivity, communication, presentations
TOTAL	20	Shared network, internet, security, backup, printing, and support

PART 2 – Enterprise Hardware Inventory

The inventory provides one primary workstation for every employee while reserving laptops for mobile work and management/technical use. A 48-port switch is selected to leave spare ports for growth and additional devices.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001–020	Desktop Computers	20	All Departments	Primary workstation for each employee
HW-021–025	Business Laptops	5	IT / Management / Shared	Mobile development, meetings, remote support, contingency
HW-026	Server	1	IT / Server Room	Central services, file services, authentication, monitoring
HW-027	Router	1	Network	Routing, WAN connectivity, NAT, basic traffic management
HW-028	Managed 48-Port Switch	1	Network	Connects wired endpoints and leaves capacity for growth
HW-029–030	Business Printers	2	HR/Finance + Shared	Document printing and scanning
HW-031–052	UPS Units	22	Workstations + Server/Network	Power protection and controlled shutdown during outages
HW-053–054	Wireless Access Points	2	Office	Reliable staff Wi-Fi and future guest Wi-Fi separation
HW-055	NAS Storage	1	IT / Server Room	Local shared storage and backup target
HW-056–057	External Backup Drives	2	IT	Offline/rotated backup copies
HW-058–077	Monitors	20	All Departments	Display for employee workstations

Hardware Justification

- Twenty desktop computers provide a dedicated primary endpoint for all 20 employees, avoiding workstation sharing.
- Five laptops provide mobility for IT, management, presentations, field support, and temporary replacement needs.
- One Ubuntu Server is sufficient for the initial small-office workload; services should be virtualized or separated logically as usage grows.
- A 48-port managed switch is preferable to a 24-port switch because 20 desktops plus servers, printers, access points, and future devices would quickly consume a smaller switch.
- Two printers reduce congestion and provide a departmental/shared printing option.
- UPS units protect critical equipment from short power interruptions and provide time for safe shutdown where appropriate.
- Two wireless access points improve coverage and allow future separation of staff and guest wireless networks.
- NAS storage plus rotated external drives provides a second local backup target and an offline copy strategy.

PART 3 – Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	Current supported release	Commercial/OEM or volume license	Primary desktop operating system with business administration features
Ubuntu Server	Current LTS release	Open source	Server operating system for internal services and infrastructure
Microsoft 365 / Office	Current supported release	Subscription	Word processing, spreadsheets, presentations, email/collaboration as licensed
Visual Studio Code	Current stable release	Free / open source components	Software development and code

			editing
Git	Current stable release	Open source	Source-code version control
GitHub Desktop	Current stable release	Free	Graphical Git/GitHub workflow for developers
VirtualBox	Current stable release	GPL/open source	Virtual machines for testing and lab environments
Google Chrome	Current stable release	Free	Web browsing and access to cloud/web applications
Microsoft Defender	Included with supported Windows edition	Included	Endpoint malware and threat protection
AnyDesk	Current supported release	Commercial license recommended for business use	Remote technical support with controlled authorization
7-Zip	Current stable release	LGPL / open source	File compression and archive management

Software Rationale

The software stack combines a supported business desktop operating system, a Linux server platform, productivity applications, developer tools, version control, browser access, endpoint protection, remote support, virtualization, and archive utilities. Licensing should be reviewed before deployment and should match the company's number of users and devices.

PART 4 – Enterprise Network Inventory

Asset ID	Equipment	Quantity	Location/Use	Purpose
NET-001	ISP Modem/ONT	1	WAN	Terminates the internet service connection
NET-002	Business Router	1	Network	Routes traffic between WAN and LAN; NAT/DHCP where appropriate
NET-003	Next-Generation Firewall	1	Network Security	Perimeter security, filtering, VPN, logging, and access control
NET-004	Managed 48-Port Gigabit Switch	1	LAN	Central wired connectivity and VLAN capability
NET-005–006	Wireless Access Point	2	Office	Staff wireless and future guest wireless
NET-007	24-Port Patch Panel	1	Server/Network Room	Organizes and terminates structured copper cabling
NET-008	CAT6 Ethernet Cable	Approx. 1,000 m	All Areas	Structured horizontal cabling with spare capacity
NET-009	RJ45 Connectors	100	IT	Cable termination and maintenance stock

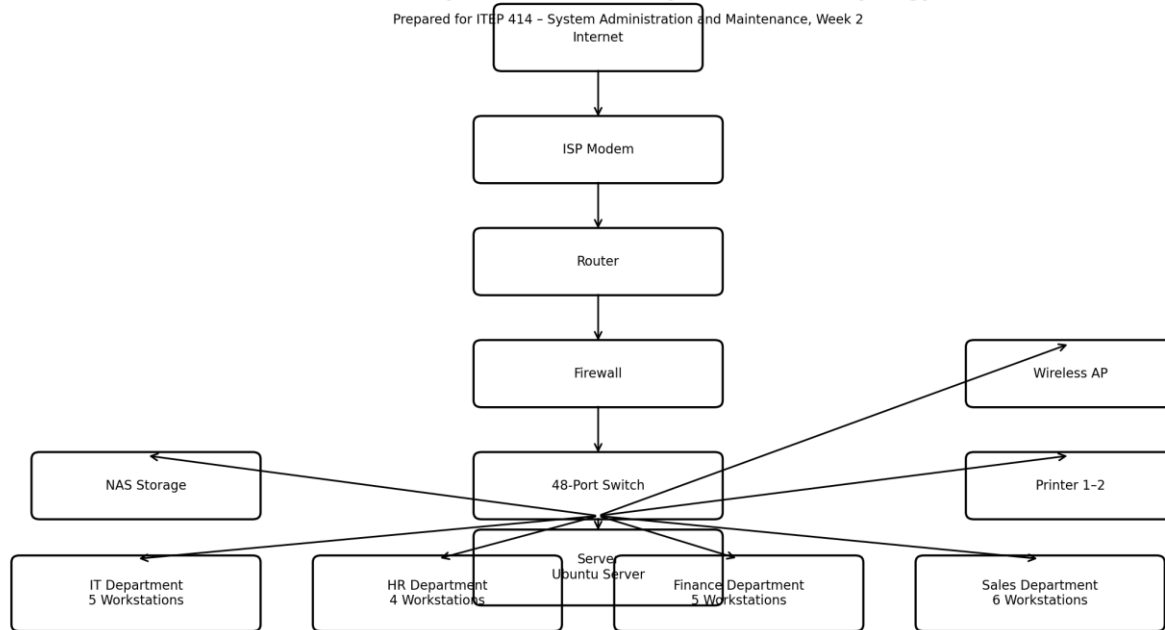
The network should use structured cabling from the patch panel to the managed switch, with labeled ports and wall outlets. The firewall should be positioned between the router/ISP edge and the internal LAN. The design can later implement VLANs for IT, HR, Finance, Sales, servers, and guest wireless.

PART 5 – Enterprise Network Diagram

The following topology satisfies the required components from the module: Internet, ISP modem, router, firewall, switch, wireless access point, server, printer, and all four departments. The logical flow is Internet → ISP Modem → Router → Firewall → Managed Switch → Internal Resources.

ABC Startup Solutions — Enterprise Network Topology

Prepared for ITBP 414 – System Administration and Maintenance, Week 2



Recommended logical segmentation: VLAN 10 – IT; VLAN 20 – HR; VLAN 30 – Finance; VLAN 40 – Sales; VLAN 50 – Servers/Infrastructure; VLAN 60 – Guest Wi-Fi. Inter-VLAN access should be controlled by firewall or Layer-3 ACL policies rather than allowing unrestricted department-to-department traffic.

Network Design Explanation

- The ISP modem/ONT provides the connection from the internet service provider.
- The router handles WAN/LAN routing functions and can provide NAT or DHCP depending on the chosen firewall architecture.
- The firewall protects the internal network by filtering inbound and outbound traffic and enforcing security policies.
- The managed switch is the central wired connectivity point and supports VLANs for logical segmentation.
- Wireless access points provide Wi-Fi without replacing the wired network; staff and guest networks should be logically separated.
- The Ubuntu Server and NAS should be placed in a restricted infrastructure/server segment.
- Printers should be centrally managed and restricted so that only authorized clients can access them.

PART 6 – System Administration Roles

6.1 Helpdesk Technician

Responsibilities

Provides first-line technical support, receives and tracks user incidents, performs basic troubleshooting, installs or configures approved software, assists with account/device issues, documents resolutions, and escalates complex incidents.

Skills

Hardware and software troubleshooting, Windows fundamentals, basic networking, customer service, ticket documentation, communication, and problem solving.

Common Tools

Ticketing/helpdesk platform, remote-support software, Windows administrative tools, device management tools, command-line utilities, and knowledge bases.

Certifications

CompTIA A+ is a strong foundational option because it validates core IT support knowledge. Vendor-specific support certifications can be added based on the organization's technology stack.

6.2 Network Administrator

Responsibilities

Designs, configures, monitors, secures, and troubleshoots switches, routers, wireless networks, firewalls, VLANs, IP addressing, DNS/DHCP, VPNs, and network performance.

Skills

TCP/IP, subnetting, routing and switching, VLANs, wireless networking, firewall concepts, network monitoring, troubleshooting, documentation, and security fundamentals.

Common Tools

Switch/router management interfaces, command-line interfaces, network monitoring platforms, packet analyzers, cable testers, IP scanning tools, and configuration backup tools.

Certifications

Cisco CCNA is a strong network-focused certification. CompTIA Network+ is another useful vendor-neutral foundation.

6.3 Linux System Administrator

Responsibilities

Deploys and maintains Linux servers, manages users and groups, storage, services, updates, permissions, logs, networking, backups, security controls, and troubleshooting.

Skills

Linux command line, shell scripting, file systems, permissions, package management, networking, systemd/services, storage, security, logging, and automation.

Common Tools

SSH, Bash, systemctl, journalctl, package managers, cron, text editors, monitoring tools, firewall utilities, and configuration-management tools.

Certifications

Red Hat Certified System Administrator (RHCSA) is a recognized practical Linux administration credential. Linux+ is another vendor-neutral option.

6.4 Cloud Administrator

Responsibilities

Manages cloud identities, compute, storage, virtual networks, security controls, monitoring, backups, resource policies, cost controls, and operational reliability.

Skills

Cloud architecture basics, networking, IAM, virtualization, storage, monitoring, automation, security, backup/recovery, and cost management.

Common Tools

Cloud portals, command-line interfaces, infrastructure-as-code tools, monitoring dashboards, identity platforms, backup services, and security consoles.

Certifications

Microsoft Certified: Azure Administrator Associate (AZ-104) is suitable for Azure administration. For AWS, the current operations-focused associate certification is AWS Certified CloudOps Engineer – Associate.

6.5 How the Four Professionals Work Together

The four roles operate as a coordinated support and infrastructure team. The Helpdesk Technician is the front line for user issues and escalates incidents that require specialized expertise. The Network Administrator maintains connectivity, switching, wireless access, routing, and network security. The Linux System Administrator maintains server operating systems, services, storage, accounts, and system-level security. The Cloud Administrator manages cloud resources, identity, virtual networking, backups, monitoring, and cloud operations. They collaborate through ticketing, documentation, change management, monitoring, and escalation procedures so that a user problem can be traced from endpoint, through network connectivity, to server or cloud services when necessary.

PART 7 – Infrastructure Recommendations

Internet Provider

Choose a business-grade fiber provider offering reliable uptime, adequate upload/download capacity, a public static IP option if required, service-level support, and fast fault escalation. For a 20-person software company, a starting business plan in the 200–500 Mbps range is reasonable, subject to actual workload and local provider availability. If the company becomes highly dependent on internet services, add a secondary ISP for failover.

Server Specifications

Use a business-class tower or rack server with at least an 8-core CPU, 32–64 GB ECC RAM, redundant or enterprise-grade storage, hardware monitoring, and a UPS. A practical initial design is mirrored SSD storage for the operating system/services plus separate storage for data. The server should run Ubuntu Server LTS and use virtualization or containers where appropriate. Capacity should be reviewed every six months.

Backup Strategy

Follow a 3-2-1 approach: keep at least three copies of important data, on two different storage types, with one copy offline or otherwise isolated from the primary environment. Use the NAS for local backups, rotate encrypted external drives for offline copies, and consider a cloud backup for critical business data. Test restoration regularly rather than assuming backups work.

Security Recommendations

Deploy a business firewall, separate network segments with VLANs, restrict administrative access, use least privilege, keep systems patched, disable unnecessary services, secure the server room, enable logging, and document incident-response and change-management procedures.

Antivirus / Endpoint Protection

Use Microsoft Defender on supported Windows endpoints and keep real-time protection, cloud-delivered protection, and security updates enabled. Centralize device/security visibility where possible and investigate alerts instead of simply dismissing them.

Password Policy

Require unique passwords, prohibit password reuse for important accounts, enable multi-factor authentication for cloud and administrative accounts, use a password manager for organizational credentials, disable shared accounts, and apply stronger controls to privileged accounts. Administrative credentials should be separate from normal user accounts.

Expansion Plan

Reserve switch ports, rack/UPS capacity, IP address space, and cable pathways. Design VLANs from the start so additional departments can be added without redesigning the entire network. When headcount grows, consider a second switch, additional access points, redundant firewall/ISP capability, a dedicated virtualization host, and cloud services for scalable workloads.

Recommended Initial Network Addressing

VLAN	Purpose	Example Subnet	Notes
10	IT	10.10.10.0/24	Developer/admin endpoints
20	HR	10.10.20.0/24	HR systems and workstations
30	Finance	10.10.30.0/24	Sensitive financial systems
40	Sales	10.10.40.0/24	Sales endpoints and applications
50	Servers/Infrastructure	10.10.50.0/24	Server, NAS, network management
60	Guest Wi-Fi	10.10.60.0/24	Internet-only access; no internal access

PART 8 – Personal Reflection

This Week 2 project taught me that system administration begins long before a computer is connected to a network. The most important lesson I learned is that an administrator must first understand the organization's people, departments, workflows, security needs, and future growth before selecting hardware or software. Creating the hardware, software, and network inventories helped me understand how business requirements are translated into technical resources. I also learned that infrastructure planning is not only about buying powerful equipment; it is about choosing equipment that is appropriate for the number of users, the work they perform, the available budget, and the expected growth of the company.

The most challenging task for me was designing the enterprise network because many components have to work together. The modem, router, firewall, switch, access points, server, printers, and employee workstations each have a different role. I also had to think about how the network could be organized into logical segments so that departments can communicate when necessary without giving every device unrestricted access to every other device. This made me appreciate the importance of network documentation, IP addressing, VLANs, access control, and clear diagrams.

Planning is important before deployment because mistakes discovered after installation can be expensive and disruptive. A good plan helps an organization estimate costs, select compatible equipment, identify security requirements, prepare backups, and establish procedures before problems occur. It also makes future troubleshooting easier because administrators have a documented picture of how the environment is supposed to work.

This project will help me become a better System Administrator because it develops both technical and professional skills. I practiced inventory management, infrastructure design, documentation, security thinking, and technical communication. In a real organization, I will need to explain technical decisions to managers and users, not just configure devices. Completing this project showed me that a good System Administrator must be organized, careful, security-conscious, and able to plan for both current requirements and future growth.